Muhammad Adam

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Research Vision

I build robust, secure, and trustworthy AI systems through adversarial robustness evaluation, red-teaming, and failure-mode characterization. My research focuses on stress-testing AI models under real-world conditions adversarial attacks, distribution shifts, and system-level vulnerabilities to uncover hidden failure modes before deployment.

Education

Carnegie Mellon University – Africa **Kigali, Rwanda**
MSc in Engineering Artificial Intelligence *Aug 2025 – Dec 2026*
- **Capstone Project:** VettedAI – Evaluation Framework for AI Candidate Scoring (team of 3–4); building validity and robustness evaluation methodology for an AI-based hiring assessment system.
- **Relevant Coursework:** Deep Learning, Adversarial ML, Hardware for AI, Mathematical Foundations of ML.

Nigerian Defence Academy **Kaduna, Nigeria**
MSc in Statistics and Data Analysis, ***CGPA: 4.52/5.00 (First Class Honour)*** *Aug 2023 – Mar 2025*
- **Thesis:** Regression Analysis of Determinants of Medical Costs – advanced training in Bayesian inference, stochastic processes, and high-dimensional probability.
- **Core:** Multivariate Analysis, Bayesian Inference, Stochastic Processes, High-Dimensional Probability.

SRM Institute of Science and Technology **Tamil Nadu, India**
BCA in Data Science, ***CGPA: 8.78/10.00 (First Class)*** *Jul 2022 – Jun 2025*
- **Thesis:** Agricultural Risk Prediction and Soil Health Monitoring Using Machine Learning.
- **Specialised Training:** Supervised/Unsupervised Learning, Deep Learning Architectures, Large-Scale Data Analytics.

International University of Africa **Khartoum, Sudan**
BSc Physics (Honours) *Aug 2014 – Sep 2019*
- **Thesis:** Numerical Solution of Deuteron Potential Depth Using Python (IOP, 2022).
- **Mathematical foundations:** numerical PDEs, statistical mechanics, matrix decompositions.

Publications

2026: K. Koech, **Muhammad Adam**, B. B. Jean Jacques, J. Barros. "Phoenix: Safe GitHub Issue Resolution via Multi-Agent LLMs." *IEEE IRAI*, Melbourne (accepted). Preprint: https://arxiv.org/abs/2606.20243

2022: **Muhammad Adam**, I.A. Alnour. "Numerical Solution of Deuteron Potential Depth Using Python." *IOP Conference Series: Materials Science and Engineering*, 1231, 012005.

Research Projects

2026 (ongoing): **MemInject: Red-Teaming Toolkit for Memory-Augmented LLM Agents.** Built a PyTorch-based library for evaluating the robustness of memory-augmented LLM agents against delayed indirect prompt injection. Implemented three injection strategies (direct write, semantic blend, retrieval bias) and a benchmark suite. Initial results on a mock memory agent show 75% trigger rate under direct injection and 67% under semantic blend; evaluation on production-grade LLM agents is in progress.

2026 (in progress): **Activation Steering for Deceptive Alignment.** Building an experimental pipeline to detect and suppress deceptive alignment in cooperative multi-agent LLM systems via representation engineering: layer-wise linear probing for a deception-related direction, direction learning, and inference-time activation steering. Pipeline validated end-to-end on a synthetic testbed; evaluation on a trained multi-agent system is the next step.

2025 (completed): **Cross-Robustness of Lightweight Vision Systems.** Systematically compared adversarial robustness (PGD, FGSM) vs. natural OOD corruptions in lightweight vision models. Quantized models lost 14% robust accuracy vs. 5% in full precision a critical insight for deploying ML in resource-constrained

environments. Code available on GitHub.

Open-Source Contributions

- **Garak (NVIDIA):** Opened a PR proposing memory-poisoning probes (MemoryInjectionProbe, DelayedTriggerProbe), covering a threat vector absent from Garak's core library.
- **meminject (Author & Maintainer):** PyTorch-based red-teaming library for memory-augmented LLM agents. Implements 3 injection strategies, includes benchmark suite, and is publicly available on GitHub.

Research Experience

CMU-Africa (Prof. Ismaila Dabo) **Kigali, Rwanda**
Engineering Independent Study *Aug 2026 – Dec 2026*

- **Project:** ML-Accelerated Prediction of Electrochemical Windows for Freeze Doping Solvent Screening in Quantum Dot Devices.
- Faculty-supervised independent research course, continuing work on representation drift and robust model design for clean-energy materials optimization.

MOSAIC Lab, CMU (Prof. Ismaila Dabo) **Kigali, Rwanda**
Research Assistant *Jun 2026 – Aug 2026*

- Developed a Python ML pipeline and a 53-candidate solvent library for predicting electrochemical windows of freeze-doping solvents in quantum dot devices.
- Built a Gaussian Process Regression surrogate model (Matérn 5/2 kernel) with calibrated uncertainty estimates, screening candidates using RDKit and partial DFT descriptors.

Carnegie Mellon University Africa **Kigali, Rwanda**
Teaching Assistant, Bridge Program *Sep 2026 – Nov 2026*

CMU-Africa **Kigali, Rwanda**
Graduate Researcher (Prof. C. Mukamakuza) *Aug 2025 – Dec 2025*
Systematically compared adversarial robustness vs. natural OOD corruptions in lightweight vision models. Key finding: quantized models lose 14% robust accuracy vs. 5% in full precision.

BMGF / NFTI **Kaduna, Nigeria**
Data Science Fellow *Sep 2023 – Mar 2024*
Selected top 1% nationwide; developed Bayesian spatio-temporal models for malaria prediction.

Honors & Awards

- Carnegie Mellon University Africa Graduate Fellowship (2025)
- BMGF Data Science Fellow – Top 1% nationwide (2023)
- e-VBAB Full Merit Scholarship (2022)
- ICICE Merit Scholarship for Physics (2015)

Technical Toolkits

Adversarial ML & Security: Red-teaming, memory/prompt injection, trigger persistence benchmarking, adversarial robustness evaluation (PGD, FGSM)

Interpretability: Linear probing, activation steering, representation-space analysis

Machine Learning: PyTorch (custom hooks, activation steering), Hugging Face Transformers, LangChain

Programming: Python (advanced), MATLAB, Git, Linux/Bash, LaTeX

Math/Stats: Bayesian inference, stochastic processes, high-dimensional probability, numerical linear algebra

Languages & Professional Memberships

- English (fluent), Arabic (fluent), Hausa (native)
- Intl. Society for Data Science & Analytics; Royal Statistical Society (e-student)